CNENGR202 Instrumentation and Control

Part 2 – L1 Capacitance based sensors

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Capacitive sensors

Aim of this part is to describe all the parts of a simple instrumentation system used to measure an unknown property such as the temperature of a room, the humidity in a greenhouse or the acceleration in a car.

Sensors

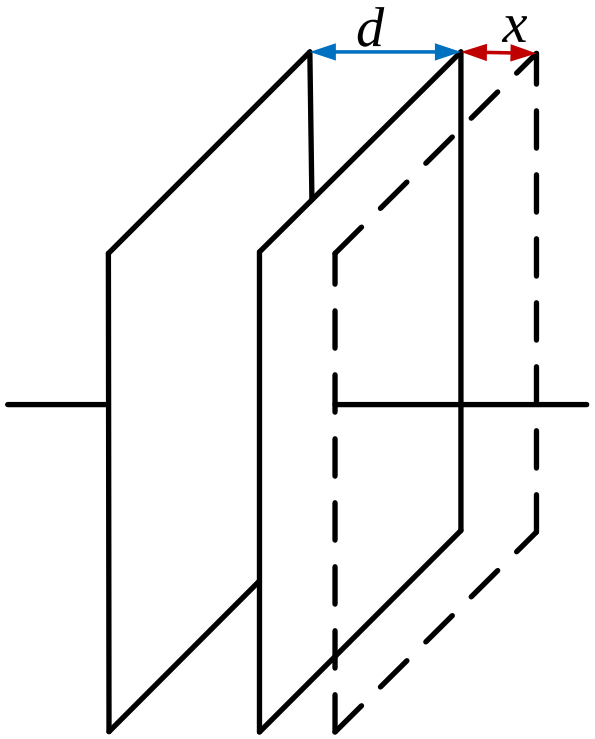
- A sensor is a device which converts one physical parameter to another

Parameter (measured)	Sensor (transducer)	Output effects
Temperature	strain gauge	Resistance
Pressure	Photo-resistor	Capacitance
Length	Thermocouple	Expansion
Light Level	Capacitor	Voltage
Force	Manometer	Current
Acceleration	Pitot tube	Pressure

Sensors - Characteristics

- **Sensitivity** of a sensor is defined as: $\frac{\Delta Output}{\Delta Input}$
- **Absolute error:** Measured value – True Value
- **Resolution:** is the smallest change that can be detected
- **Hysteresis**: Dependence of measurement on history
- **Long term instability:** magnetic domain changes, corrosion
- **Short term instability:** temperature, humidity, magnetic field
- **Calibration:** used to deduce an unknown input from a known output

Type 1: Variable gap



$$C = \frac{\varepsilon_0 \varepsilon_r A}{d + x}$$

E.G.
Capacitive
force sensor



ε_0 = permittivity of free space (F/m)

ε_r = relative permittivity

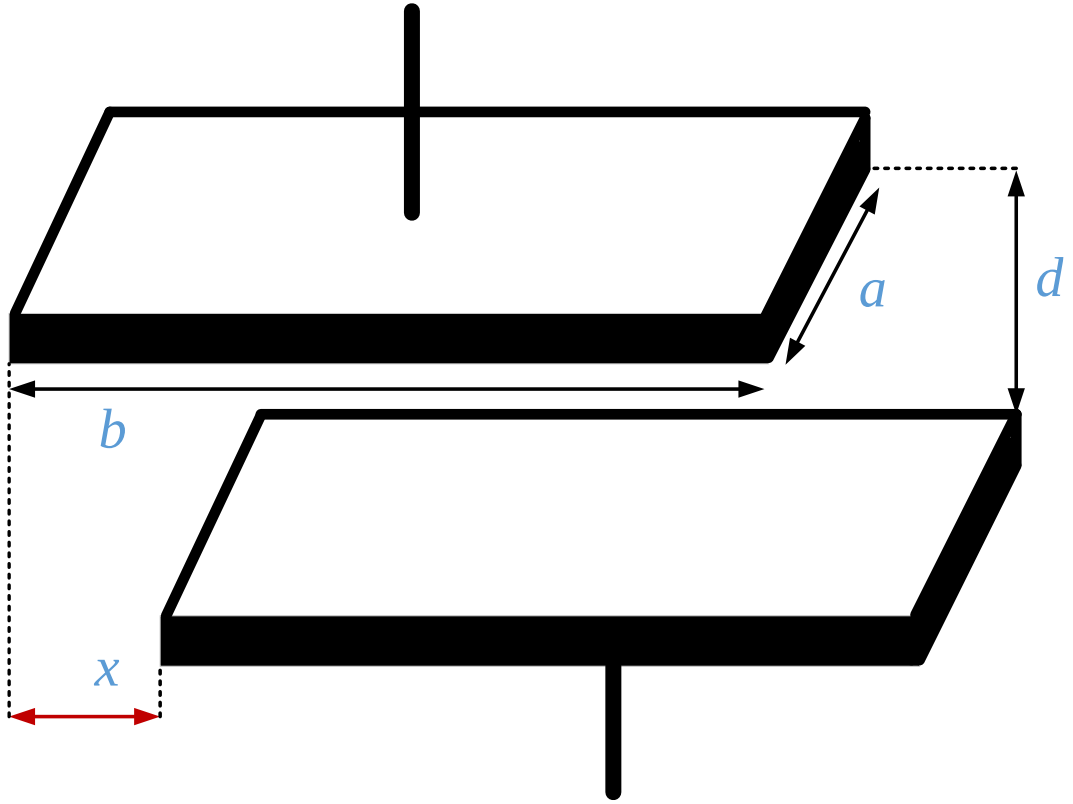
A = area of overlapping plates (m^2)

d = Separation (m)

C = capacitance (F)

x = varying distance between plates (m)

Type 2: Variable area displacement sensor



ε_0 = permittivity of free space (F/m)

ε_r = relative permittivity

b = length of capacitive plates(m)

a = width of capacitive plates(m)

x = change in position of plates(m)

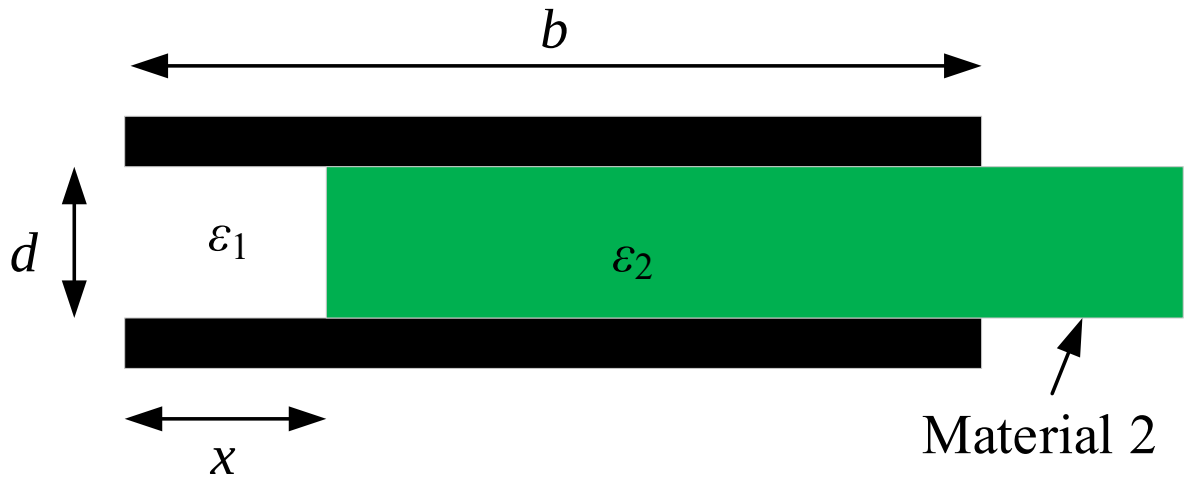
d = Separation (m)

C = capacitance (F)

$$C = \frac{\varepsilon_0 \varepsilon_r a(b-x)}{d}$$

Type 3: Variable dielectric displacement sensor

$$C = \frac{\varepsilon_0 \varepsilon_1 a x}{d} + \frac{\varepsilon_0 \varepsilon_2 a (b - x)}{d}$$



ε_0, d, C, a, b as above

ε_1, ε_2= relative permittivity of each material

x = change in position material 2

Type 4: Push-pull capacitance

differential capacitance

ε_0 : permittivity of free space (F/m)

ε_r : relative permittivity

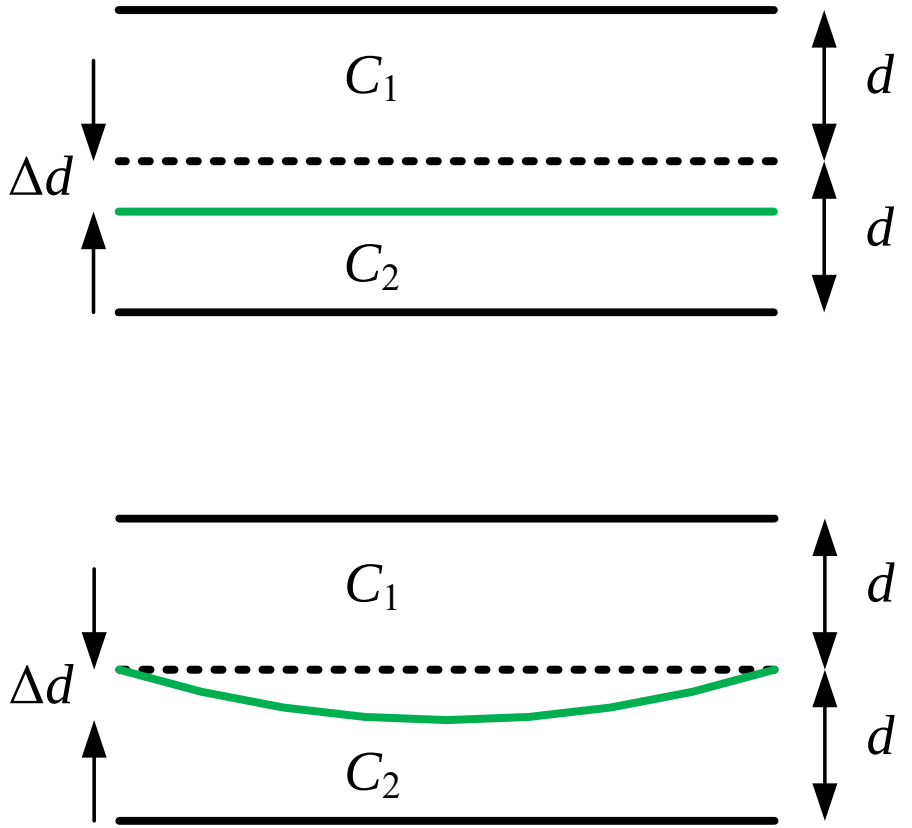
A : cross section of capacitor(m^2)

d : Separation (m)

Δd : average variation of central plate from middle (m)

C_1 : capacitance of top half (F)

C_2 : capacitance of bottom half (F)



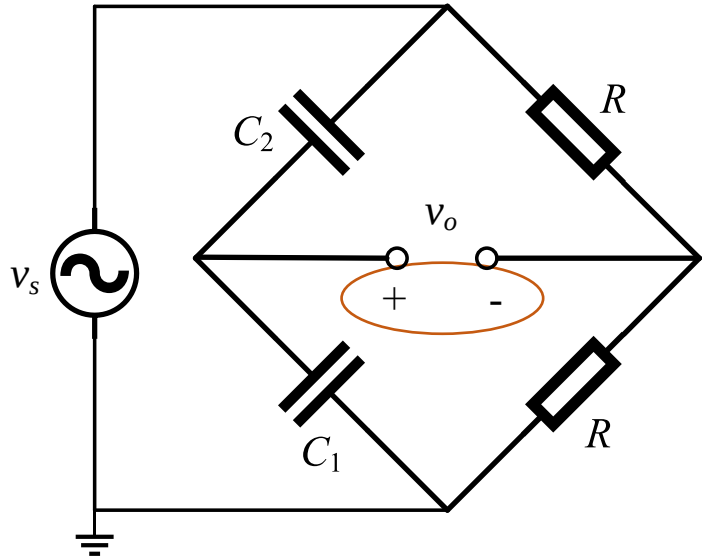
$$C = \frac{\varepsilon_0 \varepsilon_r A}{d} \qquad C_1 = \frac{\varepsilon_0 \varepsilon_r A}{d + \Delta d} \qquad C_2 = \frac{\varepsilon_0 \varepsilon_r A}{d - \Delta d}$$

Capacitive transducers interface with Wheatstone bridge - 1

$$C_1 = \frac{\varepsilon_0 \varepsilon_r A}{d + \Delta d} \qquad C_2 = \frac{\varepsilon_0 \varepsilon_r A}{d - \Delta d}$$

Transfer function: $\frac{v_o}{v_s} = ?$

Notice:
AC power supply



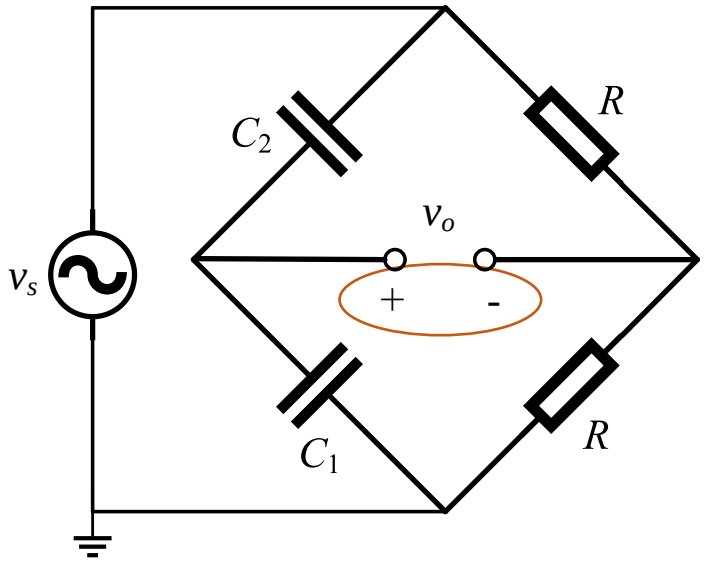
Capacitive transducers interface with Wheatstone bridge - 2

$$\frac{v_o}{v_s} = \frac{1/sC_1}{1/sC_1 + 1/sC_2} - \frac{R}{R+R}$$

$$= \frac{d+\Delta d}{(d+\Delta d)+(d-\Delta d)} - \frac{1}{2}$$

$$= \frac{d+\Delta d}{2d} - \frac{1}{2}$$

$$= \frac{\Delta d}{2d}$$



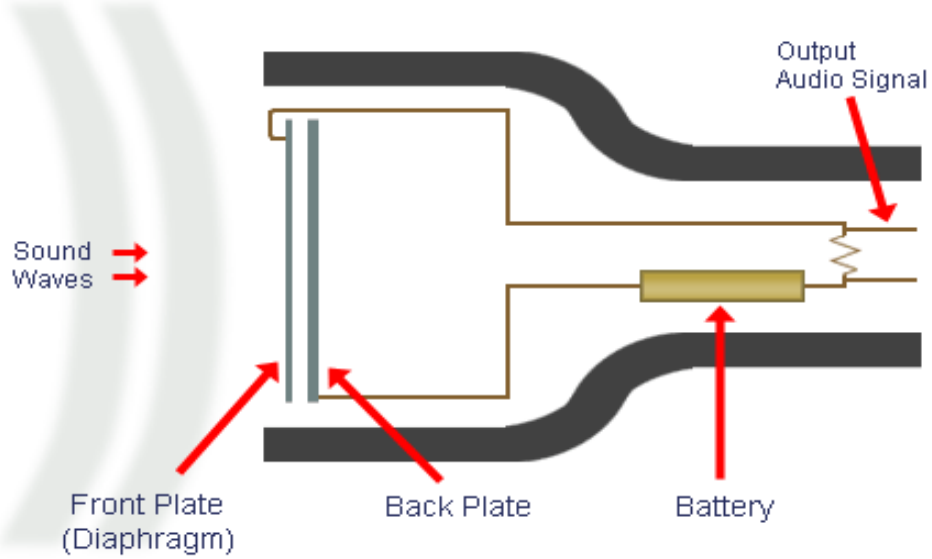
Example application of capacitance based sensor

- Condenser Microphones - sound to electrical signal



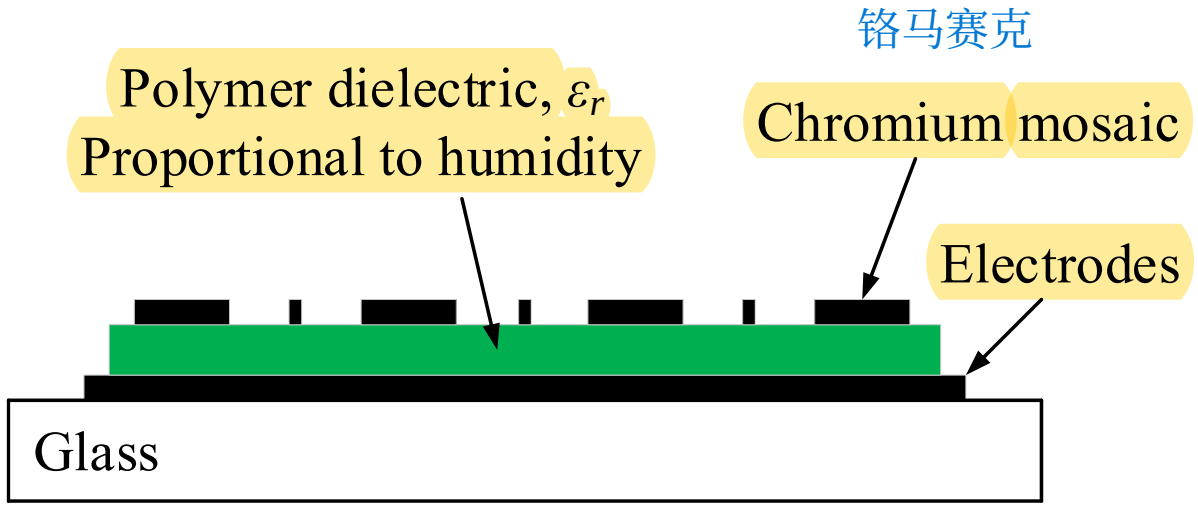


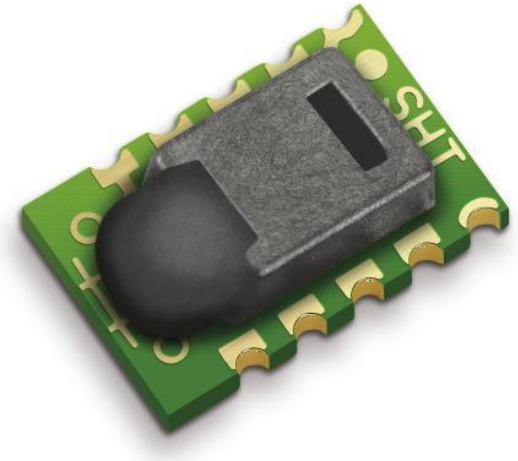
- Works by recording the capacitance between a diaphragm front plate and fixed back plate



Type 5: Variable dielectric (e.g. Humidity sensor)

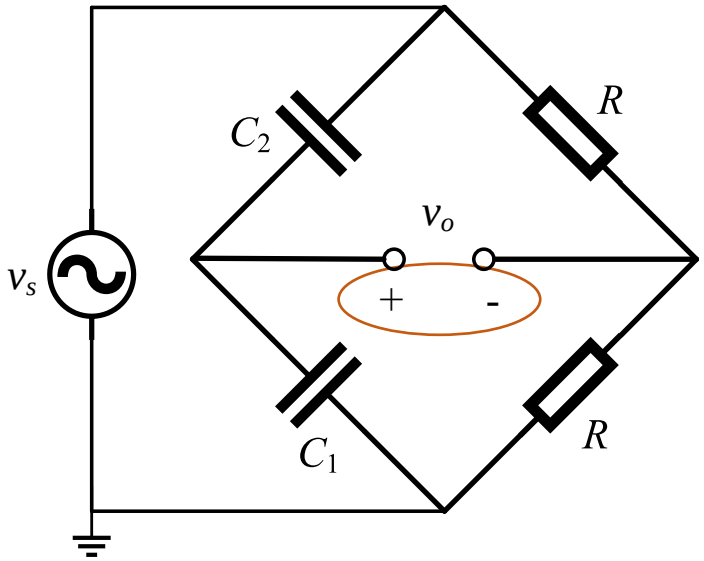
- Polymer ($\varepsilon_r \sim 2$)
- Absorbs water ($\varepsilon_r \sim 80$).
- A small absorption of water can produce a large change in capacitance.





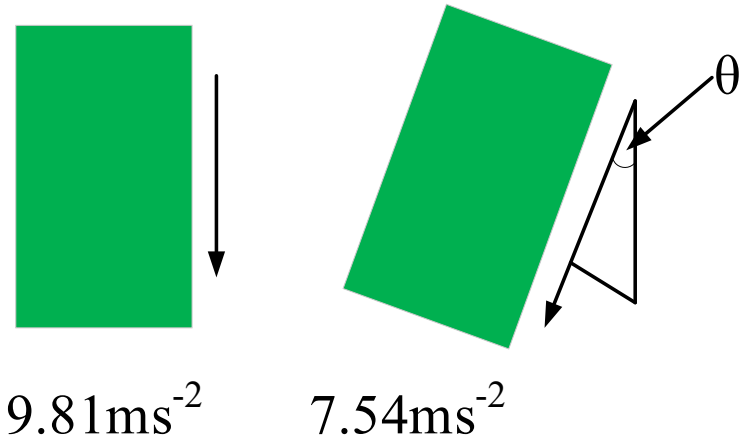
Variable dielectric interface with Wheatstone Bridge

- The capacitor is put into a bridge circuit
- This time only the one capacitance is being measured
- The fixed capacitor is used to make sure cancelling is possible



Example device using capacitance (1): Accelerometer

- Electromechanical device that measures acceleration forces which may be static (e.g gravity) or dynamic (e.g shaking).
- Can also be used as a crude gyroscope.



$$\cos\theta = \frac{7.54}{9.81} \rightarrow \theta = 39.77°$$

Example device using capacitance (2): Accelerometer

- Laptops protect hard drives from damage due to falling (detect sudden freefall, switches hard drive off thus heads don't damage drive)
- Industry standard way of detecting car crashes and deploying airbags

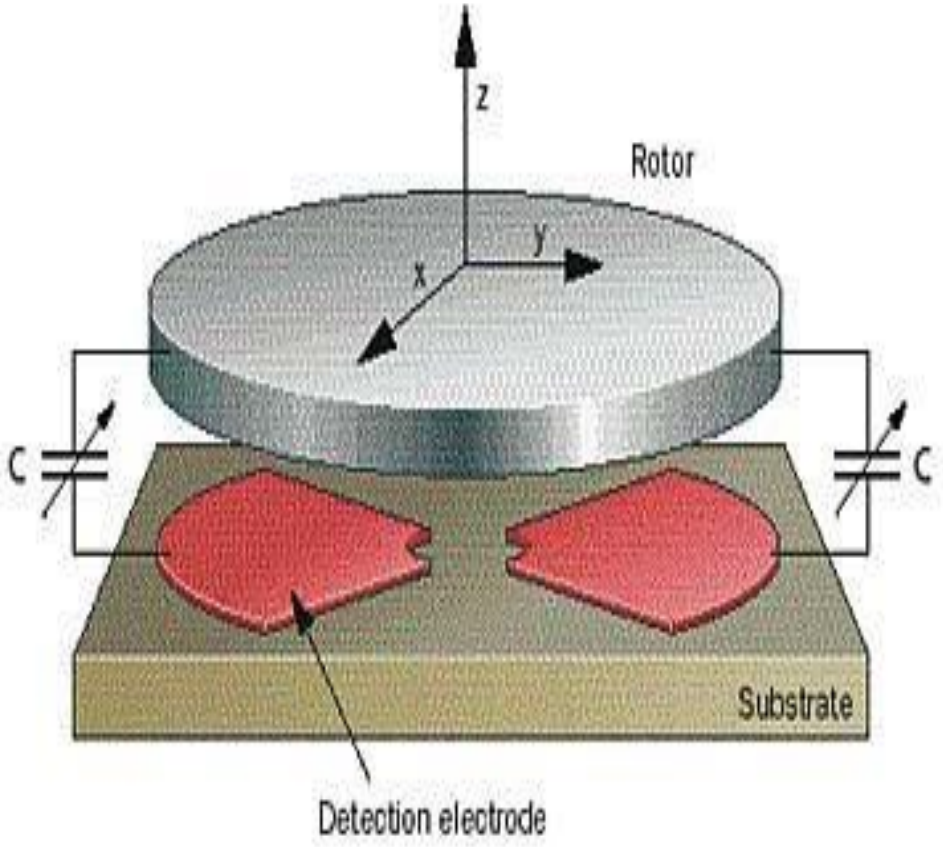
Example device using capacitance (2): Gyroscope 陀螺

A device to determine orientation and used in applications such as

- Determining spacecraft orientation
- Stabilisation based safety features on some modern cars
- Within computer game controllers
- Image stabilisation in cameras
- In robotics to determine arm orientation

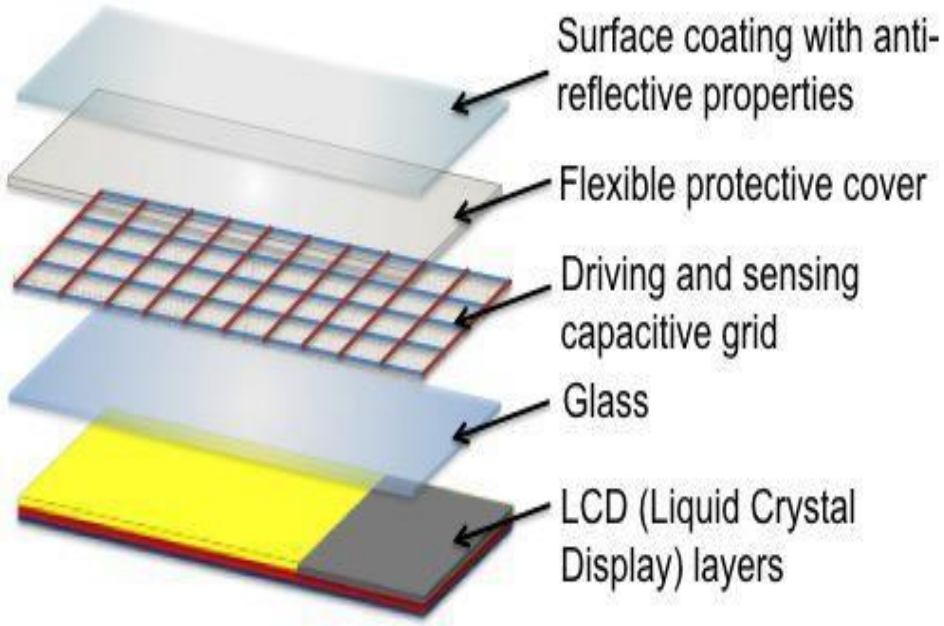
Vibrating wheel gyroscope

A wheel is driven to vibrate about its axis of symmetry, and rotation about either in-plane axis results in the substrate tilting, a change that can be detected with capacitive electrodes under the wheel.



Example device using capacitance (3): Touch screen

- Work with anything that holds an electrical charge – including human skin (hence gloves do not work unless they have conductive thread).



Example device using capacitance (3): Touch screen-1

- Capacitive touch screens are constructed from materials like indium tin oxide that store electrical charges in an electrostatic grid of micron thick wires.
- The human body has a natural capacitance, which means we are all electrically conductive and can store electric charge. When you touch the screen of your iPhone with your finger the sensing capacitance layer interacts with your natural body capacitance and this changes the electrical charge at that interaction point.

Class question 1

- Use a capacitive humidity sensor within a bridge circuit. How does the voltage output vary with capacitance and therefore humidity?

